



Genome-wide identification and classification of Lipoxygenase gene family and their roles in sorghum-aphid interaction

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Abstract

Key message This report shows detailed characterization of LOX gene family in sorghum and provides new insight of sorghum LOX genes in genetic structure and their roles in plant response to infestation by sugarcane aphids.

Abstract Lipoxygenases (LOXs) are monomeric, nonheme iron-containing dioxygenases that initiate the fatty acid oxidation pathway creating oxylipins and plant hormone jasmonate both have a key role in plant development and defense. To date, a comprehensive and systematic analysis of sorghum LOXs is still deficient. Thus, we performed a genome-wide analysis of the sorghum LOXs genome and identified nine LOXs genes. Detailed examination of protein sequences and phylogenetic analysis categorized the sorghum LOXs into two subclasses, 9-LOXs (*SbLOX1*, *SbLOX3*, *SbLOX4*, *SbLOXm*, and *SbLOXo*), 13-LOXs (*SbLOX9*, *SbLOX5*, and *SbLOX2*), and the unclassified *SbLOX8*. This classification was further supported by sequence similarity/identity matrix and subcellular localization analysis. The lipoxygenase domains, motifs, and vital amino acids were highly conserved in all sorghum LOX genes. In silico analysis of the promoter region of *SbLOXs* identified different hormones responsive cis-elements. Furthermore, to explore the roles of sorghum LOXs during sugarcane aphid feeding and exogenous MeJA application, expression analysis was conducted for all the eight LOXs in resistant (Tx2783) and susceptible (Tx7000) sorghum lines, respectively. As detailed in this report, the data generated from both genome-wide identification and expression analysis of lipoxygenase genes suggest the putative functions of two 13-LOXs (*SbLOX9* and *SbLOX5*) and three 9-LOXs (*SbLOX1*, *SbLOX3*, and *SbLOXo*) in biosynthesis of jasmonic acid, green leaf volatiles and death acids, and all of them are involved in defense-related functions in plants. Furthermore, this report represents the first genome-wide analysis of the LOX gene family in sorghum, which will facilitate future studies to characterize the roles of each individual LOXs gene in aphid resistance and defense responses to other stresses.

Keywords Aphids · Jasmonic acid · Lipoxygenases (LOXs) · Phylogenetic analysis · Sorghum · Gene expression

Introduction

Lipoxygenases (linoleate: oxygen oxidoreductase, EC 1.13.11.12) are monomeric, nonheme iron-containing dioxygenases, which are found widely in plants, animals,

and fungus (Brash 1999; Harwood 2000). In plants, LOXs are involved in several diverse functions, like seed germination, tuber development, vegetative growth, fruit ripening and most importantly in plant defense against pathogen or pests (Bailly et al 2002; Barry and Giovannoni 2007; Feussner and Wasternack 2002; Yan et al 2013). During pathogen or pest attack, LOX proteins catalyze the oxidation of polyunsaturated fatty acid (linolenic acid and α -linolenic acid) into unsaturated fatty acid hydroperoxides which subsequently produce oxylipins (Fig. 1) (Brash 1999; Feussner and Wasternack 2002; Siedow 1991). Different form of oxylipins such as jasmonates, green leaf volatiles (GLVs) and recently discovered death acid plays a major role in defense against pathogens and pests (Allmann et al 2010; Bell et al 1995; Christensen et al 2015; Shen et al 2014; Yan et al 2013). Jasmonates and GLVs

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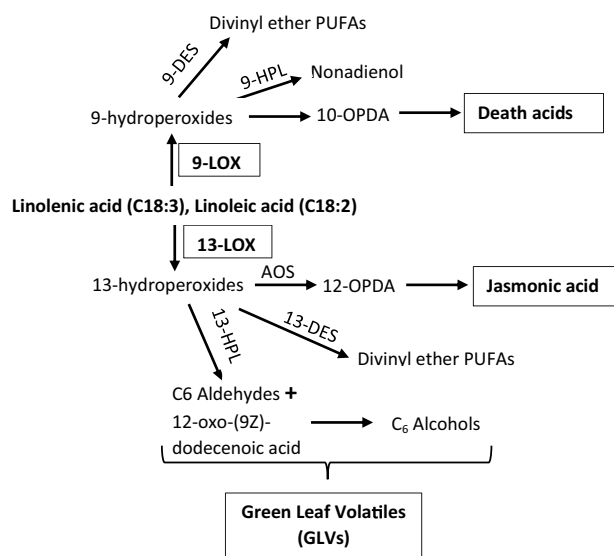


Fig. 1 13-LOX oxygenates α -linolenic acid to produce 13 hydroperoxides while 9-LOX oxygenates both linolenic acid and linoleic acid to produce 9-hydroperoxides. Both hydroperoxides serve as precursors for producing several essential plant metabolites: Death acids, green leaf volatiles and jasmonates, 10-OPDA 10-oxophytodienoic acid, 12-OPDA 12-oxophytodienoic acid, DES Divinyl ether synthase, AOS Allene oxide synthase, HPL hydroperoxidelyase Modified from Schiller et al. (2015) and Woldemariam et al. (2017)

are 13-LOX-dependent products and plays a major role in direct and indirect defenses, respectively. Jasmonic acid is an important plant stress hormone and plays a major role in defense against pathogens and herbivores (Feussner and Wasternack 2002; Zarate et al 2007). GLVs provides indirect defense by attracting the natural enemies of the herbivores (Allmann et al 2010; ul Hassan et al. 2015). Recently discovered death acids, 10-oxo-11-phytoenoic acid (10-OPEA), 10-oxo-11-phytodienoic acid (10-OPDA) and 9-hydroxy-10E, 12Z-octadecadienoic acid (9-HOD) are derived from 9-LOXs which suppressed the growth of fungi and insects (Christensen et al. 2015; 2016; Woldemariam et al 2018).

Plant LOXs are generally classified as 9-LOXs and 13-LOXs based on the specificity of their action on linoleic acids (LA) or linolenic acids (LnA) oxygenation (Feussner and Wasternack 2002; ul Hassan et al. 2015). LA or LnA is oxygenated either at C-atom 9 (9-LOXs) or C-atom 13 (13-LOXs) of the fatty acid hydrocarbon backbone (Feussner and Wasternack 2002). Comprehensively, the LOXs are classified into two sub-families, Type-I and Type-II, based on primary structure and overall sequence similarity. Type I LOXs proteins lack chloroplast transit peptides and have high sequence similarity (> 75%) within them. In contrast, Type II LOXs harbors putative chloroplast transit peptide sequence and have moderate sequence similarity (~35%) with each other. All Type-II LOXs known so far belong to

the 13-LOXs, while Type-I comprises of both 9-LOXs and 13-LOXs (Feussner and Wasternack 2002; Sarde et al 2018).

Several researches have shown LOXs as one of the important defenses signaling genes during plant-pathogen or pest interaction. In *Arabidopsis*, there are a total of six LOXs genes, four of them are 13-LOXs (*AtLOX2*, *AtLOX3*, *AtLOX4*, *AtLOX6*) and remaining two are 9-LOXs (*AtLOX1* and *AtLOX5*) (Bannenberg et al 2009). *AtLOX2* is a wound response gene (Bell et al 1995) and produces the bulk of JA during the early hour of wounding (Glauser et al 2009; Moran and Thompson 2001; Schommer et al 2008). Similarly, *AtLOX6* plays a major role in wound-induced JA formation (Feussner and Wasternack 2002). Other 13-LOXs, *AtLOX3* and *AtLOX4* play a key role in flower fertility and male sterility (Caldelari et al 2011). Among the remaining 9-LOXs, *AtLOX5* is produced by plants as a defense signaling but aphids can use it as a cue to facilitate infestation and promote fecundity (Nalam et al 2013) while *AtLOX1* is upregulated during pathogen attack and stress (Melan et al 1993). Among monocots plant, maize LOXs are most widely studied, it consists of thirteen 13-LOXs and seven 9-LOXs (Ogunola et al 2017; Woldemariam et al 2018). Among seven 9-LOXs, *ZmLOX4* functions in herbivory defense either through direct toxicity (death acids) or unknown signaling functions (Woldemariam et al 2018), in contrast, *ZmLOX3* negatively regulates resistance to fungi (Constantino et al 2013). During feeding of corn leaf aphid on maize, the higher expression of 9-LOXs (*ZmLOX1*, *ZmLOX3*, *ZmLOX4*, and *ZmLOX5*) with increased production of oxylipin was observed which might lead to aphid resistance (Tzin et al 2015). *ZmLOX1* and *ZmLOX5* expression was induced during corn borer feeding (Yang et al 2015). Among thirteen 13-LOXs, *ZmLOX12* expression showed enhance resistance to corn pathogen, *Fusarium verticillioides*. *ZmLOX13* expression was induced in early hours with subsequent higher production of JA during maize and corn leaf interaction (Tzin et al 2015). The recent finding indicates that oxylipins besides JA can have defense role in maize (Constantino et al 2013) or beneficial effects on aphid's growth and fecundity (Nalam et al 2012). The other oxylipins like death acids (Christensen et al 2015), GLVs (Christensen et al 2013), and uncharacterized compounds (mass to charge ratio 238 and 240) (Tzin et al 2015) were produced during aphid feeding suggesting its role in defense. *ZmLOX10* mediates the production of both GLVs and jasmonate as a defense compound during *Spodoptera exigua* feeding on maize (Christensen et al 2013). In rice, *OsLOX1* (9-LOX) induces resistance against brown plant hopper (BPH) (Zhou et al 2014) in contrast *OsLOX9* (13-LOXs) negatively provides resistance to BPH (Ma et al 2020; Zhou et al 2009). Several findings suggest that both 13 and 9-LOXs are mostly induced by plant species to provide a defense (producing JA, GLVs or death acids) against

herbivory but sometimes aphid can modify them or use for their benefit.

Sorghum [*Sorghum bicolor* (L) Moench] is an important cereal crop in the United States for food, fodder, and bio-fuel industry with approximately US \$2 billion economic impacts in 2015 (USDA 2017). After a major outbreak of sugarcane aphid (*Melanaphis sacchari*) (SCA) in sorghum (Beaumont, Texas, 2013), it has spread to 20 more states (Bowling et al 2016) and is now considered as a major pest of sorghum. SCA negatively affects the sorghum growth and development in two ways: by sucking the phloem sap and producing the honeydew (Singh et al 2004). Despite the commercial significance of sorghum and alarming spread of SCA in sorghum growing areas, the genetic and molecular mechanisms behind various plant responses are still unknown. For example, to develop a resistance variety against SCA, identification of genes that plays a defense role in sorghum is crucial.

Until now, little information is available about the sorghum LOX genes. LOX genes play an important role during sorghum-aphid interaction, so genome-wide study of the sorghum LOXs gene is essential. We performed comparative genomics and domain-scan analysis to identify and classify the sorghum LOX gene family. A detailed in silico study of LOX proteins was conducted to find the characteristic protein domains, sub-cellular localization and gene structure. Additionally, in silico analysis of the sorghum LOX genes identified the tandem duplication among gene pairs and cis-elements in the promoter region. Also, to find the conservation levels of sorghum LOXs phylogenetic analysis was done with six other species, including monocots and dicots. Moreover, to examine the role of sorghum LOXs in defense, expression analysis was conducted upon two treatments: SCA feeding and MeJA treatment. This research provides insight into Sorghum LOX genes and their putative functions during aphid infestation.

Materials and methods

Sequence acquisition of LOX genes

The protein sequences of Arabidopsis, rice, tomato, poplar, sweet pepper, and maize for the LOX genes, were downloaded from the NCBI (<http://www.ncbi.nlm.nih.gov/>) and phytozome <https://phytozome.jgi.doe.gov/pz/portal.html>). To identify LOX genes in sorghum two main approaches were used. First, a BLAST search was performed on the sorghum genome using the Arabidopsis LOX proteins. The protein sequences identified from the searches with E-value smaller than (E -value, 10^{-5}) were pooled and redundant sequences were removed. Second, the acquired sorghum proteins were checked for the presence of lipoxygenase domain,

LOX and PLAT/LH2 (polycystin-1, lipoxygenase, α -toxin domain or lipoxygenase homology) using the InterProScan program (<http://www.ebi.ac.uk/Tools/InterProScan/>) and Pfam database (<http://pfam.xfam.org/>) was used to confirm the presence (Quevillon et al 2005). The identified sorghum LOXs genomic DNA sequence and coding DNA sequence (CDS) were retrieved from the phytozome. These sequences were analyzed for exon/intron organization using gene structure display (GSDS) tool (Hu et al 2015).

Phylogenetic analysis of plant LOXs

The identified sorghum LOX proteins were aligned using the MUSCLE program (<https://www.ebi.ac.uk/Tools/msa/muscle/>) (Edgar 2004) with default settings. To establish the evolutionary relationship of sorghum LOX with the known LOX families of Arabidopsis, rice, poplar, tomato, sweet pepper, and maize, phylogenetic analysis was performed. A total of 81 amino acids sequence was used to construct phylogenetic tree by the maximum likelihood method with Poisson correction model using 1000 bootstrap values (Felsenstein 1985; Zuckerkandl and Pauling 1965) in MEGAX program (Kumar et al 2018). Additionally, these nine sorghum lipoxygenase sequences were also run through the sequence identity and similarity (SIAS) program (<http://imed.med.ucm.es/Tools/sias.html>) to calculate the identity and similarity matrix.

Sequence analysis and identification of conserved motifs

Structural motif annotation for all the nine LOXs genes were performed using the MEME program (<http://meme-suite.org/tools/meme>). The maximum number of motifs was set at 20. We performed manual observation on sorghum LOX alignments to identify the conserved sequences and pivotal amino acids. The online tool Weblogo (Crooks et al 2004) was used to generate the sequence logos of conserved regions present in sorghum LOX proteins. Subcellular localization of the proteins was identified through the CELLO2GO (Yu et al 2014). Further to see the presence of chloroplast transit peptide (cTP), all the sorghum LOX were scanned with ChloroP 1.1 (Emanuelsson et al 1999) and TargetP-2.0 (<http://cbs.dtu.dk/services/TargetP/>). Isoelectric point (PI) and molecular weight for each LOX were predicted through the ExPasy bioinformatics program (<https://www.expasy.org/>).

Gene duplication and calculation of Ka/Ks values

Sorghum LOXs gene chromosomal locations were obtained from the Phytozome database. The two types of duplication: segmental duplication and tandem duplication gene pairs

were identified as previously (Upadhyay and Mattoo 2018). Segmental repeat gene pairs consist of the 90% sequence similarity, while five or fewer genes within the 100 kb region are considered as tandem repeats (Upadhyay and Mattoo 2018). The identified segmental pair amino acid was first aligned using the Clustal O, and the aligned sequences with their respective cDNA sequences were further aligned using the PLA2NAL program (Goldman and Yang 1994). This program uses the CODEML program of PAML to calculate synonymous (Ks) and nonsynonymous (Ka) substitution rates.

Prediction of cis-regulatory elements in promoter sequences

The 2000 bp promoter sequence upstream to ATG for SbLOX gene members was obtained from the phytozome. These obtained upstream segments were analyzed using the PlantCARE promoter analysis tool (Lescot et al 2002). Thus, obtained various cis-elements were used to construct the heatmap through gplots package in R-software (Team RC 2013).

Plant growth conditions, SCA rearing, and MeJA bioassays

Seeds from resistant (Tx2783) and susceptible (Tx7000) sorghum (*Sorghum bicolor*) lines were treated with a fungicide (Captan 50 W, Bonham, TX) and planted separately in a plastic pot (7.5 cm in diameter; 7.1 cm in height) containing potting compost (Sungro Professional Growing Mix, Agawam, MA). Seedlings were grown in greenhouse at constant temperature (28 ± 2 °C) and 60% relative humidity under constant photoperiod of 14 h-light/10 h-dark. Sugarcane aphid colonies were cultured on seedlings of susceptible sorghum line (Tx7000) in pots (15 cm × 14 cm) fitted with 12.5 cm × 30 cm cylinders of Lexan™ (SABIC Polymer-shapes, Tulsa, OK) sleeve cages with netting cloth covering on the top. Sorghum seedlings of 10–12 days old (3–4 leaf stage) were infested with 20 adults of apterous sugarcane aphid to the adaxial surface of the first leaf. To restrict aphid escape and avoid infestation from unwanted pests/pathogens, each infested plant was covered with a transparent cylindrical cage with nylon mesh on the top. Control plants, not infested with aphids, were also covered in cages (4.3 cm in diameter and 40 cm in length). For JA treatment both lines were sprayed with 100 μM of MeJA (treatment) or mock-treated with water (control), both mixed with 0.1% of Tween20. The JA and water solutions were sprinkled on the whole plants until the runoff and the plant seedlings were covered with the cages immediately. To confirm differential responses to aphid between the two genotypes (Tx7000 and Tx2783), each seedling was infested with 20 adults of

sugarcane aphids as mention early, then aphid count and plant damage ratings (chlorosis/necrosis) were recorded at 1, 2, 3, 6, 9 and 12 dpi, respectively. Ten independent plants of each genotype were included and plant damage scores were recorded using a scale of 0–5, with 0 being no damage, 1 damage < 20%, 2 damage 21–40%, 3 damage 41–60%, 4 damage 61–80%, 5 damage > 80% and 6 being dead plant.

RNA extraction, cDNA synthesis, and quantitative RT-PCR

Quantitative RT-PCR (qRT-PCR) was used to estimate the relative expression of sorghum LOXs in response to aphid infestation and JA treatment. Samples, first two true leaves and stem below the second leaf, were harvested at 0, 6, 12, 24 and 48 h post infestation and were immediately deep frozen in liquid nitrogen and stored at -80 °C. For both aphid infestation and JA treatments, controls (uninfested with SCA and untreated with JA) samples were collected at each time point to eliminate the circadian rhythm effect on gene expression. Three biological replicates were harvested and analyzed for each time point. A Trizol reagent (Invitrogen, Carlsbad, CA) was used to extract the total RNA from each sample, and then it was treated with DNase (Turbo DNA free kit, ThermoFisher, Waltham, MA). A total of 1.5 μg of RNA was reverse transcribed using the GoScript reverse transcriptase kit (Promega, Madison, WI) and cDNA was diluted to four-fold before using for the qRT-PCR reaction. Primers were designed using the IDT DNA program (<https://www.idtdna.com/PrimerQuest/Home/Index>), which are listed in Table S1. A sorghum β-tubulin gene (Sobic.002g350400) was used as the internal control as described previously (Li and Huang 2017). qRT-PCR was performed on a Bio-Rad icycler thermal cycler (Bio-Rad Laboratories, Inc., Hercules, CA, USA) using the iTaq™ universal SYBR® green super-mix (Bio-Rad Laboratories, Inc.). The qRT-PCR reaction was performed in a total volume of 10 μl, containing 1 μl of cDNA, 0.4 μl (10 μM) of reverse and forward primers, five microliters of SYBR green master mix and 3.2 μl of ddH₂O using the following conditions: one cycle at 95 °C for 3 min, 40 cycles at 95 °C for 10 s and 55 °C for 30 s, followed by one cycle each of one min at 95 °C and 55 °C. The final melting curve was of 81 cycles at 55 °C for 30 s.

Statistical analysis

The relative expression level of each gene was calculated using the $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen 2001) and the data presented are the average of three biological and two technical replicates. For the aphid infestation, aphid count and MeJA experiments, student's t-test was used to estimate the significant differences (* $P < 0.05$, ** $P < 0.01$, and *** $P < 0.01$).

Results and discussion

Identification of the lipoxygenase gene family in sorghum

We performed a genome-wide analysis of the lipoxygenase family in sorghum using two approaches: DNA sequence homology searching and scanning of the proteome for the presence of "lipoxygenase" and "PLAT/LH2" domains. From the first approach we identified fourteen lipoxygenase genes in sorghum. The second approach of scanning the presence of lipoxygenase domain and PLAT/LH2 narrowed the sorghum lipoxygenases to nine (Table 1). The decrease in the lipoxygenase genes is due to the missing of one of the domains, lipoxygenase or PLAT/LH2 (Table S1). The true lipoxygenase sequence should contain both PLAT/LH2 and lipoxygenase domains (Chen et al 2015; Song et al 2016). The number of *SbLOX* genes is close to that in *Arabidopsis* (6) (Umate 2011) but less than its close monocots, such as rice (14) (Upadhyay and Mattoo 2018) and maize (13) (Ogunola et al 2017). However, the number is higher than in olive (4) (Padilla et al 2009, 2012) and kiwifruit (6) (Zhang et al 2006) and over than in poplar (20) and tomato (14) (Upadhyay and Mattoo 2018). The lipoxygenase family gene numbers are diverse in several plant species, suggesting that this gene has not been conserved during evolution, though there is similarity in biochemical functions across different plant species (Feussner and Wasternack 2002).

Table 1 summarizes the characteristics of the sorghum lipoxygenase gene family, including accession numbers and gene features. Among them, two newly identified lipoxygenase genes *Sobic.001G125800* and *Sobic.001G125700* were assigned with the names as *SbLOXm* and *SbLOXo*, respectively, in this study. At the genomic level sequence ranges from 3575 to 7119, while in coding DNA sequence (CDS) it ranges from 2575 to 2775 with exon counts ranging from four to nine. The length of *SbLOX* proteins varied between amino-acid residues (aa) 866 and 924, the predicted molecular weight ranged from 96 to 103 kDa. Surprisingly, sorghum LOX proteins showed clear variation in PI values, ranging from 5.77 to 8.68, suggesting these proteins will require different ionic strength and/or pH range for optimal activity (Upadhyay and Mattoo 2018).

Phylogeny and structural analysis of sorghum lipoxygenase genes

In this study, the MEGAX program was used to determine the evolutionary relationship and to predict the

Table 1 Structural features of lipoxygenase genes in Sorghum

Name	Gene ID	Chromosome	exon count	Orientation	Location coordinates	Genomic sequence	Transcript	CDS	Amino acid	PI	MW (kDa)
<i>SbLOX1</i>	<i>Sobic.001G125900</i>	1	7	minus	9,916,344..9921119	4776	3015	2601	866	6.01	96.66
<i>SbLOX2</i>	<i>Sobic.001G483400</i>	1	9	minus	75,467,368..75473615	6248	3083	2775	924	8.60	102.92
<i>SbLOX3</i>	<i>Sobic.003G385500</i>	3	7	minus	69,796,879..69800694	3816	2984	2607	868	6.13	97.54
<i>SbLOX4</i>	<i>Sobic.003G385900</i>	3	6	minus	69,827,192..69830766	3575	2851	2622	873	5.77	98.25
<i>SbLOX5</i>	<i>Sobic.006G095600</i>	6	7	plus	46,566,240..46571064	4825	3769	2775	924	6.36	103.48
<i>SbLOX8</i>	<i>Sobic.008G191000</i>	8	6	plus	62,495,797..62499472	3676	3194	2574	857	8.68	96.73
<i>SbLOX9</i>	<i>Sobic.004G078600</i>	4	4	minus	6,461,422..6465436	4015	3209	2754	917	6.62	103.71
<i>SbLOXm</i>	<i>Sobic.001G125800</i>	1	9	minus	9,902,653..9909771	7119	3227	2667	888	6.26	100.5
<i>SbLOXo</i>	<i>Sobic.001G125700</i>	1	8	minus	9,896,488..9900431	3944	3195	2682	893	7.5	101.45

CDS Coding DNA sequence, PI isoelectric point, MW Molecular Weight

*LOX genes having only PLAT domain or LOX domain are listed in Supplementary Table S1

classification of sorghum LOXs. First, 72 known lipoxxygenase protein sequences were collected from six different plant species comprising of both monocots and dicots. Then a phylogenetic tree was developed for all identified LOX sequences, categorizing them into three clades, 9-LOXs, 13-LOXs, and an unknown group, which is in line with the previous categorization (Fig. 2) (Sarde et al 2018; Upadhyay and Mattoo 2018). In sorghum, among nine LOXs, three are 13-LOX (*SbLOX9*, *SbLOX5*, and *SbLOX2*) and five are 9-LOX (*SbLOX1*, *SbLOX3*, *SbLOX4*, *SbLOXm*, and *SbLOXo*) and one is unknown (*SbLOX8*). Surprisingly, there is only four lipoxxygenase gene in unknown clades, one in each of the four species, rice, maize, poplar, and sorghum. Meanwhile, a detailed examination of this 9-LOX showed two exclusive clusters of monocots and dicots, which suggest that these genes might have evolved after the separation of monocots and dicots. In contrast, the 13-LOX clade displayed the altered distribution of monocots and dicots, which implies that these genes may have already existed before the divergence of monocots and dicots (Chen et al 2015).

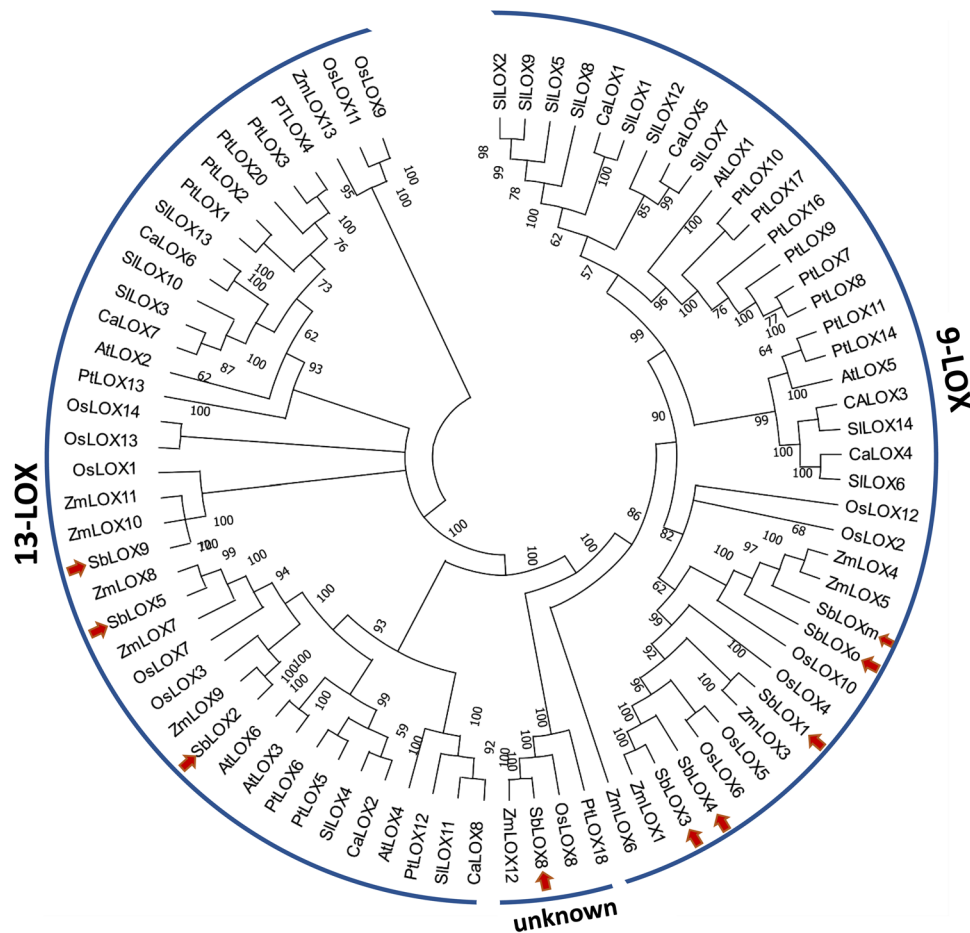
Further a separate phylogenetic tree of sorghum LOX proteins and intron/exon organization of coding sequence

of each *SbLOX* gene was constructed (Fig. 3) to study the structural diversity of sorghum LOX genes. The sorghum LOX gene were classified into two sub-families 9-LOX and 13-LOX. The sorghum genes are mapped to chromosomes (Chr) 1, 3, 4, 6 and 8, out of ten total sorghum Chr. Sorghum Chr 1 harbors four *LOX*s, Chr 3 harbors two *LOX*s, and other Chr 4,6 and 8 harbors one *LOX* each. Most members of *SbLOX* gene family were found to share exon number ranging from seven to nine. The 9-LOXs, shared six to nine introns in their open reading frame while 13-LOXs shared four to nine (Fig. 3). All of the sorghum LOXs have 5' and 3' untranslated regions (UTRs) except *SbLOX2* (missing 3'UTRs). Three 9-LOXs, *SbLOX3*, *SbLOX4* and *SbLOX1* showed similar exon arrangements suggesting the origin from the same ancestral gene and possible duplication within these genes (Chen et al 2015).

Sequence analysis confirms the phylogenetic classification of sorghum LOXs

The lipoxxygenase family of sorghum shows at the protein sequence level with variable sequence identity and similarities between each other (Table S2). LOX can also be

Fig. 2 Phylogenetic analysis of the LOX gene family. The circular tree represents the evolutionary relationship between sorghum LOX protein family and those in Arabidopsis, rice, maize, tomato, sweet pepper and poplar. The tree was generated by MEGAX using Maximum Likelihood method based on the Poisson correction model using 1000 bootstrap values (Zuckerkanndl and Pauling, 1965; Felsenstein, 1985). The analysis involved 81 amino acid sequences and they are divided into three classes: 9-LOX, 13-LOX and unknown. Species abbreviations used for phylogeny are as follows. *At* Arabidopsis thaliana, *Sl* Solanum lycopersicum, *Os* Oryza sativa, *Zm* Zea Mays, *Sb* Sorghum bicolor, *Pt* Populus trichocarpa, *Ca* Capsicum annum



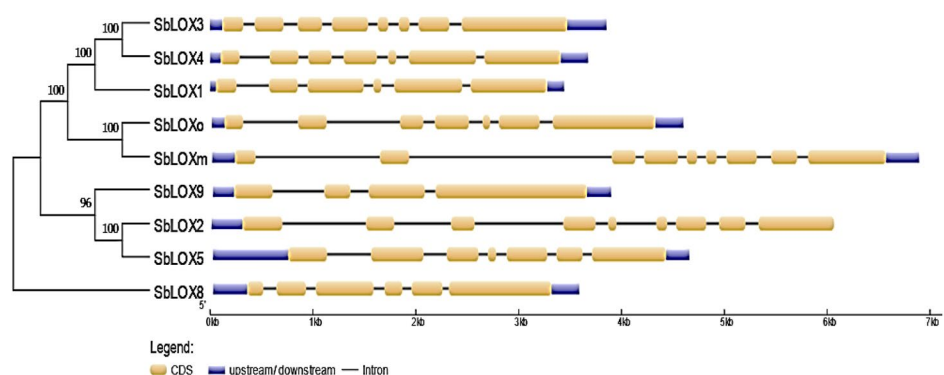


Fig. 3 Phylogenetic relationship among sorghum LOX proteins and gene structures of sorghum LOX genes. The tree was constructed by the Neighbor-Joining (NJ) method with 1000 bootstrap replicates. The schematic genomic structure for each LOX gene was gener-

ated using gene structure display server (GSDS). The size of Coding DNA sequence (CDS, yellow), introns (black lines) and untranslated regions (UTRs, purple) are proportional to their sequence lengths

classified as Type I and Type II, based on the extent of sequence similarity and the presence/absence of chloroplast transit peptide sequence (Feussner and Wasternack 2002). Type-I LOXs have a high sequence similarity (> 75%) and lacks chloroplast-transit peptide; in contrast, Type-II LOXs has moderate sequence similarity (~35%) with the presence of chloroplast-transit peptide. The sorghum LOXs, SbLOX1, SbLOX3, SbLOX4, SbLOXm, and SbLOXo, exhibit high sequence similarity (61 to 76%) and identity (63 to 84%) between each other in comparison with other LOXs (Table S3 and Table S4). In contrast, SbLOX9, SbLOX5, and SbLOX2 show moderate sequence similarity and identity between each other except for *SbLOX5* and *SbLOX2*. Similar sequence similarity and identity was observed in sweet pepper in both 9 and 13-LOXs (Sarde et al 2018). Furthermore, chloroplast transit peptide is present in all the Type-II LOXs (SbLOX9, SbLOX2, SbLOX5) (Table S4), which suggest their localization in chloroplast. Both, sequence similarity and sub-cellular localization classified sorghum LOX into two types: Type-I (SbLOX1, SbLOX3, SbLOX4, SbLOXm, and SbLOXo) and Type-II (SbLOX9, SbLOX5, and SbLOX2).

Alternatively, plant LOXs can also be classified based on the positional specificity of the specific amino acids (Phe, His or Val) in a protein sequence (Feussner and Wasternack 2002). The presence of Phe608/His608 or Val608 residue in the bottom area of substrate-binding pockets predicts LOX activity as 13 or 9-LOX, respectively (Fig. S1). Multiple sequence alignment performed in MUSCLE showed the presence of valine residue in all members of *SbLOX1*, *SbLOX3*, *SbLOX4*, *SbLOXm*, and *SbLOXo*, classifying them as 9-LOX. Similarly, the presence of phenylalanine is in *SbLOX9*, *SbLOX5*, and *SbLOX2*, classifying them as 13-LOX (Fig S1). However, there is a presence of methionine in SbLOX8 classifying them in the unknown group.

This agrees with the hypothesis proposed by Feussner and Wasternack 2002, that all the Type-II LOXs identified so far belong to the 13-LOX subfamily. Therefore, classifying sorghum lipoxygenase using these two methods provides consensus on the distribution of sorghum LOXs into two groups, thus supporting our methods and predictions. Additionally, it also suggests that both approaches can be applied to classify LOX genes in plants in the future.

Furthermore, phylogenetic analysis and molecular characterization at the DNA sequence level offer the powerful tools to predict the regiospecificity as well as sub-cellular localizations of plant LOXs whereas substrate specificity prediction focuses their function based on additional biochemical data (Feussner and Wasternack 2002). The phylogenetic tree developed above reveals that 13-LOX clade represent a major group of which more than one LOX gene can be involved in JA biosynthesis (Wasternack and Parthier 1997). Almost all plant species has at least two or more 13-LOXs genes, such as sorghum (3), *Arabidopsis* (4), sweet pepper (4), maize(5), which are expressed differentially upon wounding and are possibly involved in JA biosynthesis (Heitz et al 1997; Royo et al 1996).

High conservation in LOX motifs and pivotal amino acids

Conserved motifs of LOXs were identified using the MEME suite. Sorghum LOX proteins were found to possess 20 distinct motifs, fifteen of which are shared among all the LOX proteins. Similar number of motifs (20) was found in tomato and poplar lipoxygenase, among them 17 and 14 are shared among all the LOX proteins of tomato and poplar respectively (Chen et al 2015; Upadhyay and Mattoo 2018). These motifs and its location vary between the subfamilies but remain almost similar within the LOX family. Motif analysis

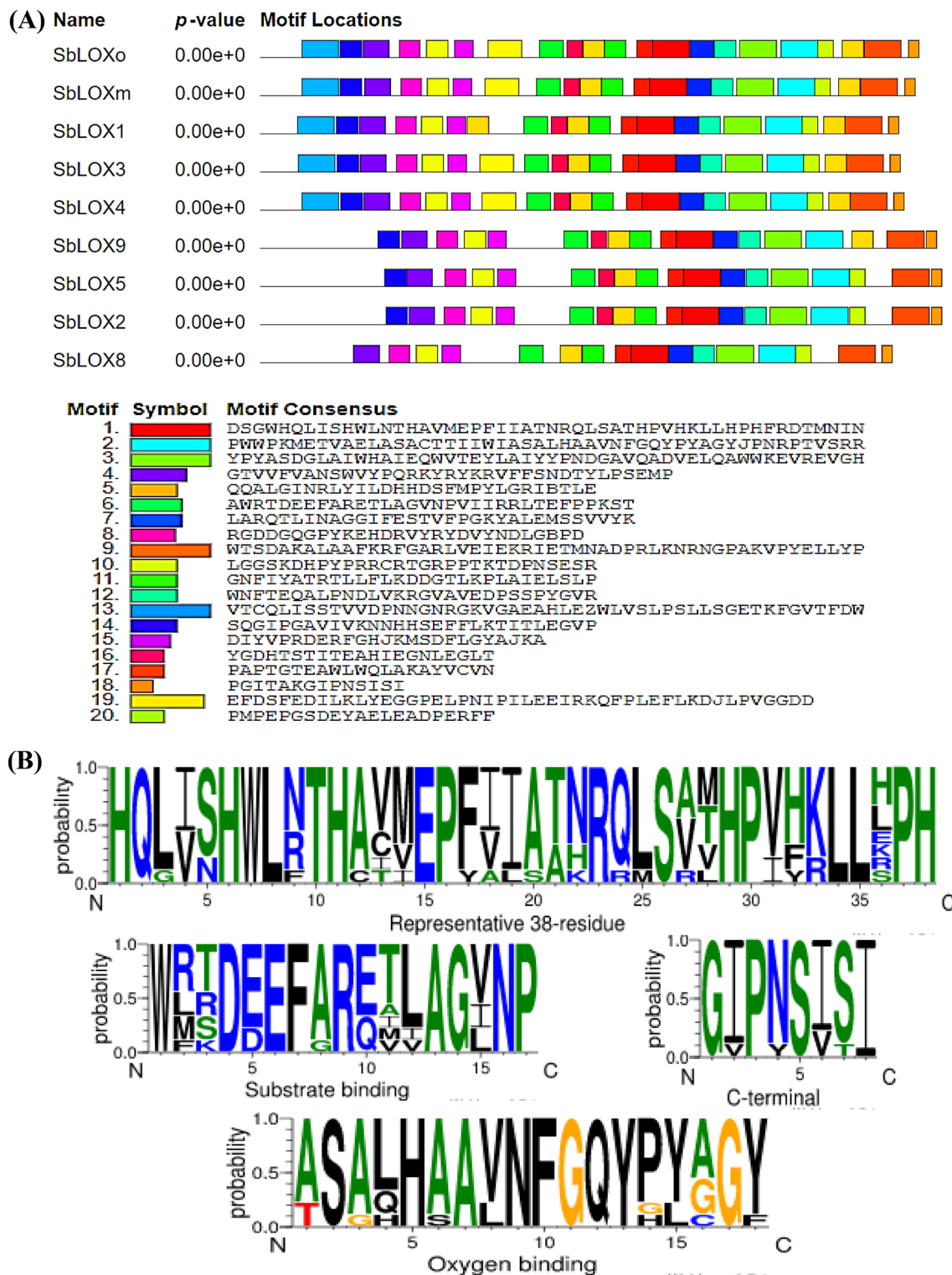


Fig. 4 **a** Schematic representation of the conserved motifs in sorghum LOX proteins. Conserved motifs (a minimum of 20) and their alignment among LOX proteins was elucidated by using MEME tool. Most of the protein possess similar domains revealing their conserveness. Each colored box represents a motif in the protein, with the consensus motif sequence below in the index. **b** Sequence alignments of

38-residue, substrate binding, C-terminal and oxygen binding motifs in LOX proteins. The sequence logo for each substrate was created using all the nine sorghum LOX sequences. In each stack, y-axis represents the sequence conservation at that position. Each residue height represents the relative distribution frequency of this residue in all the nine sorghum LOX motifs

reveals that all the 9-LOX sub-family have 20 conserved motifs, while 13-LOXs have only 17, and the motifs were similar within the sub-family (Fig. 4a). Motif 12 (in a red box) consists of the representative 38 amino acid residues and contains a histidine signature, His-(X)4-His-(X)4-His-(X)17-His-(X)8-His. This structure is highly conserved in all the nine lipoxygenases (Fig. 4a, b) and presumably plays an important role in enzyme stability and efficiency (Fig S1 pink highlight) (Steczko et al 1992; Zhang et al 2006). More importantly, sequence change in these residues can result in a significant effect on enzyme activity or efficiency of lipoxygenases (Chen et al 2015). Besides, other motifs like substrate binding, oxygen binding, and the C-terminal are also conserved among all sorghum LOXs (Fig. 4b, Fig S1). Further among these conserved amino acids, the three histidine residues (His499, His504, His609), Asn694 and Ile839 are considered as vital amino acids for binding to non-heme iron (Fig S1 red box) (Feussner and Wasternack 2002; Padilla et al 2012). The alignment in Fig S1 proved the conservation of these five amino acids in the sorghum LOXs. Altogether, conservation of above-mentioned amino acids and motifs of lipoxygenase in sorghum LOXs implies that the function of sorghum LOXs are conserved to their similar homologs of other plant species.

Gene duplication and estimation of Ka/Ks values

Phylogenetic analysis and chromosomal position information were analyzed to determine the gene duplication. There are two main ways of gene duplication, tandem or segmental, that increase the number of genes in a family. For segmental duplication gene pairs, *SbLOXo-SbLOXm* (75.29% similar) and *SbLOX3-SbLOX4* (75.59%) came close but did not match up the requirement of 90% similarity for segmental repeats consideration (Sharp et al 2005). Based on the phylogenetic and chromosomal position two tandem duplication was found among *SbLOX3-SbLOX4* in Chr 3 and *SbLOX1, SbLOXo and SbLOXm* in Chr1. To further explore the driving force of LOX evolution, we estimated the substitution rate ratios of nonsynonymous (Ka)

versus synonymous substitution (Ks) (Table 2). This Ka/Ks ratio indicates the selection pressure acting on a gene or its region, the ratio less than 0.5 suggests purifying selection for both duplicates in a gene pair (Chen et al 2015). A summary of Ka/Ks values are shown in Table 2, all the values in four gene pairs are less than 0.5. Similar result was observed in the tomato (Upadhyay and Mattoo 2018) and poplar (Chen et al 2015). The result indicates that all the LOX gene pairs have evolved through purifying selection pressure and these genes are slowly evolving at the protein level (Chen et al 2015; Upadhyay and Mattoo 2018).

Identification of cis-elements for hormonal and defense-related cues in *SbLOX* gene promoters based on in-silico analysis

In silico analysis of sorghum LOXs gene promoters was carried out using the Plant Care promoter databases (Lescot et al 2002). The analysis revealed that almost all the *SbLOXs* gene promoters were decorated with different hormone-responsive cis-elements (Fig. 5; Table S5). The two cis-acting regulatory elements (CGTCA-motif and TGACG-motif) involved in the MeJA response were present in all the *SbLOX* genes. The two cis-acting elements (TC-rich and TC-rich repeats) involved in defense and stress responsiveness were present in all the 13-LOXs (*SbLOX2, SbLOX5, and SbLOX9*) and *SbLOX4*. Similarly, stress and defense inducible element promoter CCAAT-box was found in all the 13-LOXs and two 9-LOXs genes (*SbLOX3 and SbLOXo*). Salicylic acid responsive elements (TCA-element) were found in *SbLOX2, SbLOX3, SbLOX8, and SbLOXm*. Others hormones-related cis-elements like ABRE responsible for abscisic acid is found in all *SbLOXs*, while gibberellin responsiveness elements were found only

Table 2 Analysis of tandem duplication events of sorghum LOX gene pairs

Duplicated genes	Ka	Ks	Ka/Ks	Purifying selection
<i>SbLOX3-SbLOX4</i>	0.0872	0.8130	0.1073	Yes
<i>SbLOXm-SbLOXo</i>	0.1101	0.8525	0.1291	Yes
<i>SbLOXm-SbLOX1</i>	0.2071	8.7704	0.0236	Yes
<i>SbLOXo-SbLOX1</i>	0.2116	10.6200	0.0199	Yes

Ka nonsynonymous substitutions, Ks synonymous substitutions, Ka/Ks < 0.5 is considered as purifying selection for tomato lipoxygenases (Chen et al 2015)

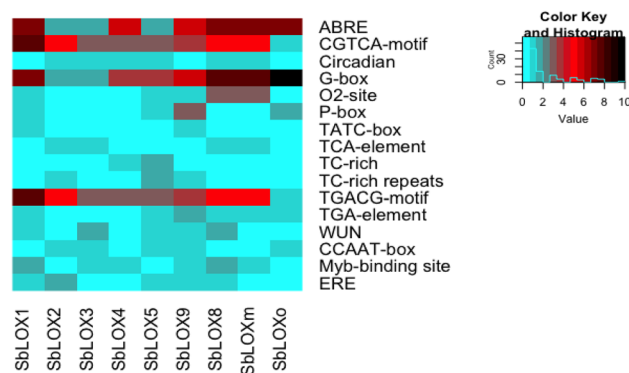


Fig. 5 Heatmap representing the distribution of cis-elements in the promoter regions (2000 bp upstream) of sorghum LOX genes. The scale (color key and histogram scale), x-axis (value) represents the number of cis-elements in each *SbLOXs* genes (1: sky blue, 5: red, 10: black), and y-axis (count) sum the number of distinct color (each motif) of overall *SbLOXs* genes

in *SbLOX1* and *SbLOX9*. Importantly wound-related motif (WUN) was found in *SbLOX1*, *SbLOX3*, *SbLOX5*, *SbLOX8*, and *SbLOX9*, and metabolism regulated motif (O2-site) was present in *SbLOX1*, *SbLOX5*, *SbLOX9*, *SbLOX8*, and *SbLOXm*. Other stress and defense inducible element Myb-binding site (MBS) was found in all the genes except in *SbLOX2*, *SbLOX5*, and *SbLOXo*. Other cis-elements present in *SbLOXs* are the circadian motif, light responsiveness (G-box), Myb-binding site (MBS), and ethylene rich motif (ERE). Overall, heatmap (Fig. 5) exhibited the higher number of MeJA and abscisic acid responsive cis-elements in *SbLOX1*, *SbLOX4*, *SbLOX5*, *SbLOX9*, *SbLOX8* and *SbLOXm* genes.

Response of Tx2783 and Tx7000 to SCA infestation

When the two sorghum cultivars, Tx2783 and Tx7000, were infested by the sugarcane aphid (SCA), the two genotypes showed differential responses to aphid feeding. Following aphid infestation, Tx2783 showed no apparent damage by the aphids, but Tx7000 was confirmed as susceptible to SCA. According to the damage ratings, Tx2783 showed a strong resistance with average damage score at 1 or less, whereas Tx7000 showed severe damage at 12dpi with a rating of 5 (Table S6). On the other hand, the resistant genotype Tx2783 exhibited adverse effect on aphid development and regeneration when compared to susceptible Tx7000. Based on number of aphid per plant, the rate of aphid regeneration was significantly reduced on Tx2783 in comparison to Tx7000 in 6-, 9- and 12-days post infestation (dpi) (Table S6 and Fig S2).

Expression pattern of 13-LOX genes upon SCA feeding and MeJA application

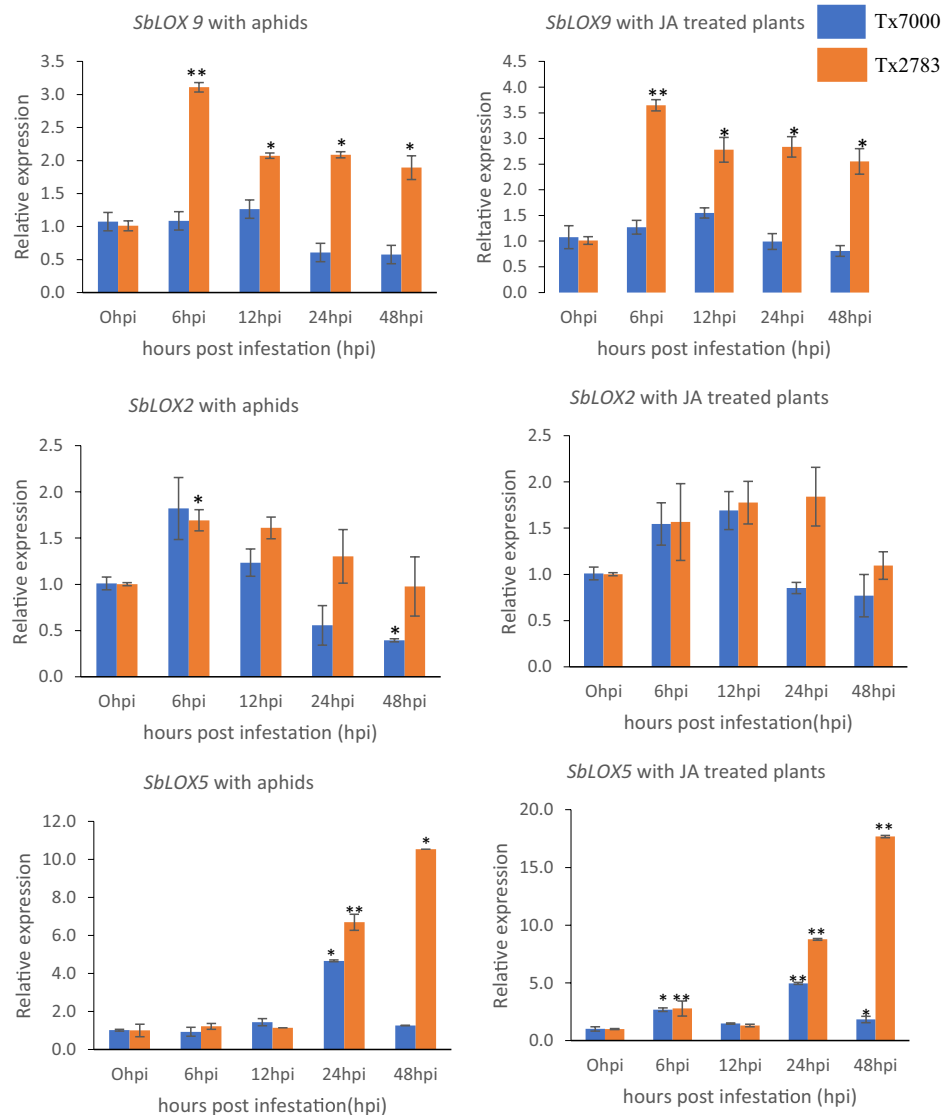
Expression pattern and dynamics of 13-LOX genes over time upon SCA feeding and MeJA application were analyzed using qRT-PCR. 13-LOX genes catalyze the α -linolenic acid (18:3), the first step in JA biosynthesis, to synthesize the Jasmonic acid, a precursor to JA-dependent defense signaling. In sorghum, *SbLOX9*, *SbLOX5* and *SbLOX2* are characterized as 13-LOX based on amino acid position and phylogenetic analysis. Upon SCA feeding, two 13-LOXs (*SbLOX9* and *SbLOX5*) were upregulated at different time points (Fig. 6). *SbLOX9* was induced in the first 6 h of infestation in resistant lines and were slightly lower in consequent hours but remained two-fold higher throughout the time points. This result was consistent with the report by Tzin et al 2015 on maize and corn leaf aphid infestation. *ZmLOX13*, closest to *SbLOX9*, was increased six-fold at early hours of infestation and was slightly decreased to two to three-fold and remained constant until 48 h. Similarly, in *Arabidopsis* *LOX2* gene, closest to *SbLOX9*, was responsible for the

bulk of JA formation in the first hour of wounding (Glauser et al 2009; Schommer, et al 2008). In another example, during *Bemisia tabaci* infestation, mutant tomato increased the production of *LOX4* in the first 12 h post infestation (hpi) resulting in JA production subsequently (Zhang et al 2018). In this study, *SbLOX5* initially showed no expression in both the susceptible and resistant lines but increased significantly in both at 24 hpi and 48 hpi (Fig. 6). On the other hand, the expression of *SbLOX5* was significantly higher in resistant lines at both 24 hpi and 48 hpi when compared with the susceptible line. Our result was consistent with the report by (Chang et al 2012), in which they found that the LOX gene (13-LOX) was highly expressed in sorghum resistant lines in comparison with susceptible lines during greenbug feeding at 48 hpi. Similarly, *CaLOX2*, like *SbLOX5*, was also highly induced in thrips and pepper interaction (Sardec et al 2018). The upregulation of *SbLOX9* and *SbLOX5* might play a key role in the JA synthesis and JA-dependent defense in the resistant line. In contrast, *SbLOX2* showed a weak expression (below two-fold) in both the lines throughout the time points (Fig. 6). A similar expression pattern was observed in thrips and pepper interaction where *CaLOX8*, closest to *SbLOX2*, showed similar low expression (Sardec et al 2018).

Several lines of evidence have shown that JA can autoregulate its pathway. Treatment of *A. thaliana* leaves exogenously with jasmonates induce JA biosynthesis endogenously (Reymond et al 2000). Interestingly, those plants treated with MeJA also tend to have higher LOX gene activity (Wang and Wu 2005). Upon exogenous JA application, *SbLOX9* and *SbLOX5* showed significant induction at various time points (Fig. 6). More importantly *SbLOX9* was strongly induced in resistant line after 6 h, then decreased at 12 hpi and remained constant afterward. Similarly, *SbLOX5* also showed a significant increase in expression at 24 hpi in resistant lines. In contrast, *SbLOX2* gene expression was weakly induced, showing below two-fold in both lines throughout all time points (Fig. 6). The increase in LOX gene expression following MeJA treatment was also noted in tomato (Upadhyay and Mattoo 2018) pepper (Sardec et al 2018) and poplar (Chen et al 2015). The upregulation patterns of three 13-LOX genes was similar in sorghum in response to attack by aphid. The significant LOX gene expression in resistant line following either aphid infestation or JA treatment suggests that these genes might be potential candidate genes involved in JA biosynthesis and JA dependent defense. The increase in the LOX genes with subsequent JA production has been reported in maize-aphid interaction (Tzin et al 2015) and tomato-whitefly interaction (Zhang et al 2018). They have reported increased JA production in infested plants during the first 48 h.

As reported in several plant species, the elevated level of JA was associated with insect resistance (Ellis et al 2002; Mewis et al 2006; Shivaji et al 2010). Previous results

Fig. 6 Expression of sorghum 13-LOXs genes in response to sugarcane aphid infestation and MeJA treatment. Twenty young aphids were placed in the first true leaf of 3–4 leaf stage (12–14 days old) sorghum seedlings. The gene expression for each *SbLOX* was normalized using the housekeeping gene, sorghum β -tubulin gene (Sobic.002g350400) at each time point. Relative expression for each *SbLOX* was estimated by comparing it to the control samples of the same time point. Error bars in each bar represent $\pm$ standard error ($n=3$). The asterisk in each bar indicate the significant differences (Student's t-test) to corresponding control samples for the same time point, * $P<0.05$, ** $P<0.01$. The bars without asterisk, are non-significant ($P>0.05$)



showed that tomato mutant can slower nymphal growth of *B. tabaci* by activating JA dependent defense. It was further supported by spraying JA in mutant tomato which reduced the nymphal population significantly (Zhang et al 2018). However, *B. tabaci* adults were able to suppress JA-dependent defense in tomato and continue their survival and reproduction. Thus, increase in 13-LOX genes and subsequent JA production does not guarantee the JA dependent defense signaling. A possible explanation for the limited expression of JA-related defense genes may be due to aphid-manipulated SA-JA crosstalk (Thaler et al 2012; Zarate et al 2007). Aphids can manipulate JA signaling by increasing JAZ genes that acts negatively to regulate the JA signaling pathway. In sorghum, greenbug marginally and temporarily induced JA regulated defense genes (Zhu-Salzman et al 2004). Moran and Thompson 2001 reported that green peach aphid infestation on Arabidopsis upregulated SA dependent

pathway and reduced the JA-related defense in host plants. Additional research is needed to confirm whether induced JA during aphid feeding can activate JA dependent defense or not in sorghum.

Comparatively, 13-LOX derived green leaf volatiles (GLVs) are less explored areas in both monocots and dicots that are interacted with insect herbivory. GLVs inhibit the growth and multiplication of plant pathogens, bacteria, viruses, and fungi, while during herbivory infestation it acts as a cue to attract parasitic or parasitoid enemies (ul Hassan et al. 2015). In maize *ZmLOX10* (13-LOXs) mediates production of both GLVs and jasmonate to defend against herbivory (Christensen et al 2013). When *S. exigua* attacked maize, the *ZmLOX10* mutants showed reduced production of GLVs throughout the time points, while the wild type (WT) plants showed significantly higher production of GLVs at 4 hpi. *SbLOX9*, a homolog closest to *ZmLOX10*, was highly

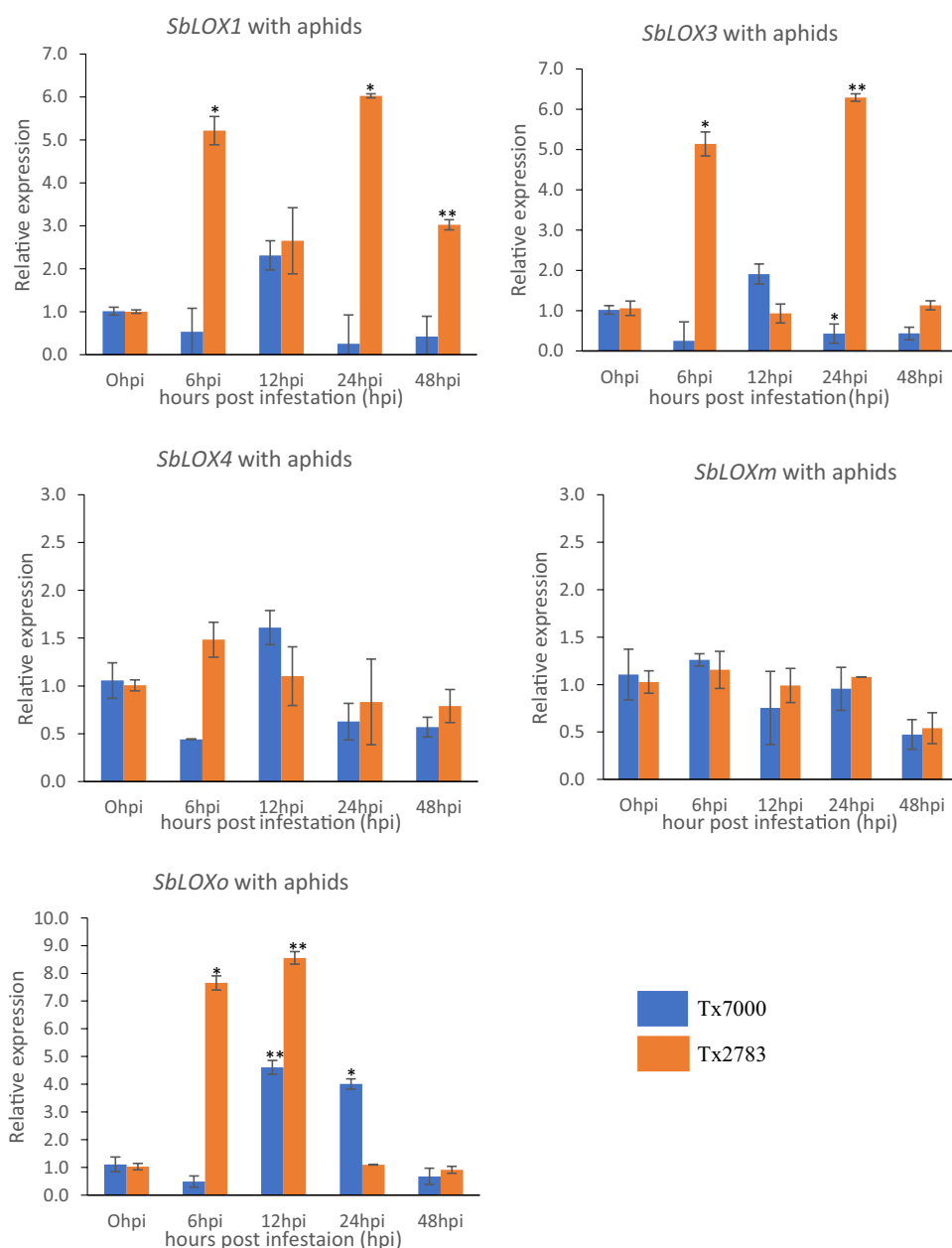
expressed at the first 6 hpi in resistant lines, then slightly lower at 12 hpi and remained two-fold higher throughout the time points (Fig. 6); thus, this gene might be involved in the production of GLVs.

Expression pattern of 9-LOXs in sorghum during SCA feeding

During aphid feeding, 9-LOXs genes oxidizes 18:3 α -linolenic acid and 18:2 α -linoleic acid to produce a variety of oxylipins, which plays some roles in plant growth, development and stress response. These oxylipins, besides JA, have a defense role in maize or beneficial effects on

aphids growth and fecundity directly (Constantino et al 2013; Nalam et al 2012). In sorghum, *SbLOX1*, *SbLOX3*, *SbLOX4*, *SbLOXm*, *SbLOXo* are characterized as 9-LOX genes based on amino acid position and phylogenetic analysis. Upon SCA feeding, three sorghum 9-LOXs (*SbLOX1*, *SbLOX3*, and *SbLOXo*) were highly upregulated at different time points. Both *SbLOX1* and *SbLOX3* were highly expressed in early hours then slightly decreased and further increased to six-fold at 24 hpi in the resistant line (Fig. 7). This result was consistent with the previous report on maize and corn leaf aphid infestation (Tzin et al 2015). *ZmLOX3* (closest to *SbLOX1*) and *ZmLOX1* (closest to *SbLOX3*) showed 2 to 4-fold increases in the first 6 hpi,

Fig. 7 Expression of sorghum 9-LOXs genes in response to sugarcane aphid infestation. Twenty young aphids were placed in the first true leaf of 3–4 leaf stage sorghum seedlings. The gene expression for each *SbLOX* was normalized using the housekeeping gene, sorghum β -tubulin gene (Sobic.002g350400) at each time point. Relative expression for each *SbLOX* was estimated by comparing it to the control samples of the same time point. Error bars in each bar represent $\pm$ standard error ($n=3$). The asterisk in each bar indicate the significant differences (Student's t-test) to corresponding control samples for the same time point, * $P < 0.05$, ** $P < 0.01$. The bars without asterisk, are non-significant ($P > 0.05$)



then a slight decrease in 24 hpi in *ZmLOX3* only. Further *ZmLOX3* and *ZmLOX1* were increased six-fold and 20-fold in 48 hpi, respectively. The up- and down-regulation of these two genes during aphid feeding suggests that there might be a continuous arms race going on between aphid and plant species to dominate each other. Remarkably, in both resistant and susceptible lines two 9-LOXs, *SbLOX4* and *SbLOXm*, showed consistent expression in all the time points (0.5 to 1.5-fold) (Fig. 7), suggesting no role of these genes in oxylipin biosynthesis. However, *SbLOXo* was highly expressed in early hours (eight-fold), 6 hpi and 12 h, and faded away in later hours (Fig. 7). A similar result was seen in *ZmLOX5*, closest to *SbLOXo*, which showed around a four fold increase in early hours (Tzin et al 2015). Remarkably, the susceptible line also showed a significant increase in expression in 12 hpi constantly in all the three genes, *SbLOX1*, *SbLOX3*, and *SbLOXo*, but comparatively less than resistant lines (Fig. 7). The expression of 9-LOXs was much higher in comparison with 13-LOXs in early hours, which is consistent with the previous findings of *S. exigua* infestation on maize (Woldemariam et al 2018). In resistant lines, the increase in the production of oxylipins induces the phytoalexin-deficient4 (PAD4) gene, which controls the severity of green peach aphid (GPA) infestation in Arabidopsis. However, GPA has evolved itself to utilize the plant LOX5-synthesized oxylipins as stimulatory cues to colonize itself successfully, which explains the increased production of oxylipin in susceptible lines (Nalam et al 2012).

For the first time, our study has identified and characterized nine lipoxygenase genes in sorghum. Through phylogenetic and sequence analysis these LOXs genes were classified into 9-LOXs (*SbLOX1*, *SbLOX3*, *SbLOX4*, *SbLOXm*, and *SbLOXo*) and 13-LOXs (*SbLOX9*, *SbLOX5*, and *SbLOX2*) with their putative functions. Detail examination of these sorghum LOXs sequence and motif analysis displayed highly conserved lipoxygenase characteristics in all of the sorghum LOXs genes. The presence of conventional motif (Histidine signature), His-(X)4-His-(X)4-His-(X)17-His-(X)8-His, indicates the role of this motif in family-specific functions. The quantitative RT-PCR revealed the different patterns of expression on 9-LOXs and 13-LOXs during aphid feeding and MeJA treatment. This result may indicate the role of these genes in specific stress-related functions. The higher expression of two 13-LOXs (*SbLOX9* and *SbLOX5*) indicates the putative functions of these genes in the biosynthesis of JA and GLVs. Similarly, a higher expression of three 9-LOXs (*SbLOX1*, *SbLOX3*, and *SbLOXo*) indicates the putative functions in synthesis of death acids. Further research is needed to confirm the JA or GLVs dependent defense signaling and death acids dependent defense during SCA feeding in sorghum.

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Authors Contribution Conceived and designed the experiment: KS, SP and YH. Performed the experiments: KS and SP. Analyzed the data: KS and YH. Wrote the manuscript: KS and YH.

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References

- Allmann S, Halitschke R, Schuurink RC, Baldwin IT (2010) Oxylipin channelling in *Nicotiana attenuata*: lipoxygenase 2 supplies substrates for green leaf volatile production. *Plant, Cell Environ* 33:2028–2040
- Bailly C, Bogatek-Leszczynska R, Côme D, Corbineau F (2002) Changes in activities of antioxidant enzymes and lipoxygenase during growth of sunflower seedlings from seeds of different vigour. *Seed Science Research* 12:47
- Bannenber G, Martínez M, Hamberg M, Castresana C (2009) Diversity of the enzymatic activity in the lipoxygenase gene family of *Arabidopsis thaliana*. *Lipids* 44:85
- Barry CS, Giovannoni JJ (2007) Ethylene and fruit ripening. *J Plant Growth Regul* 26:143
- Bell E, Creelman RA, Mullet JE (1995) A chloroplast lipoxygenase is required for wound-induced jasmonic acid accumulation in Arabidopsis. *Proc Natl Acad Sci* 92:8675–8679
- Bowling RD et al (2016) Sugarcane aphid (Hemiptera: Aphididae): a new pest on sorghum in North America. *Journal of Integrated Pest Management* 7:12
- Brash AR (1999) Lipoxygenases: occurrence, functions, catalysis, and acquisition of substrate. *J Biol Chem* 274:23679–23682
- Caldelari D, Wang G, Farmer EE, Dong X (2011) Arabidopsis *lox3* *lox4* double mutants are male sterile and defective in global proliferative arrest. *Plant Mol Biol* 75:25–33
- Chang J, Duan X, Cui J, Xue W, Zhang Q (2012) Differential molecular responses of aphid-sensitive and aphid-resistant sorghum lines to aphid infestation. *Arthropod-Plant Interactions* 6:113–120
- Chen Z, Chen X, Yan H, Li W, Li Y, Cai R, Xiang Y (2015) The lipoxygenase gene family in poplar: identification, classification, and expression in response to MeJA treatment. *PLoS ONE* 10:e0125526
- Christensen SA et al. (2015) Maize death acids, 9-lipoxygenase-derived cyclopentenone (a) nones, display activity as cytotoxic phytoalexins and transcriptional mediators. *Proceedings of the National Academy of Sciences* 112:11407–11412
- Christensen SA et al (2013) The maize lipoxygenase, Zm LOX 10, mediates green leaf volatile, jasmonate and herbivore-induced plant volatile production for defense against insect attack. *Plant J* 74:59–73
- Christensen SA, Huffaker A, Hunter CT, Alborn HT, Schmelz EA (2016) A maize death acid, 10-oxo-11-phytoenoic acid, is the predominant cyclopentenone signal present during multiple stress and developmental conditions. *Plant Signaling Behavior* 11:e1120395

- Constantino N et al (2013) Root-expressed maize lipoxygenase 3 negatively regulates induced systemic resistance to *Colletotrichum graminicola* in shoots. *Frontiers in Plant Science* 4:510
- Crooks GE, Hon G, Chandonia J-M, Brenner SE (2004) WebLogo: a sequence logo generator. *Genome Res* 14:1188–1190
- Edgar RC (2004) MUSCLE: a multiple sequence alignment method with reduced time and space complexity. *BMC Bioinformatics* 5:113
- Ellis C, Karafyllidis I, Turner JG (2002) Constitutive activation of jasmonate signaling in an *Arabidopsis* mutant correlates with enhanced resistance to *Erysiphe cichoracearum*, *Pseudomonas syringae*, and *Myzus persicae* *Molecular Plant-Microbe Interactions* 15:1025–1030
- Emanuelsson O, Nielsen H, Heijne GV (1999) ChloroP, a neural network-based method for predicting chloroplast transit peptides and their cleavage sites. *Protein Sci* 8:978–984
- Felsenstein J (1985) Confidence limits on phylogenies: an approach using the bootstrap. *Evolution* 39:783–791
- Feussner I, Wasternack C (2002) The lipoxygenase pathway. *Annu Rev Plant Biol* 53:275–297
- Glauser G, Dubugnon L, Mousavi SA, Rudaz S, Wolfender J-L, Farmer EE (2009) Velocity estimates for signal propagation leading to systemic jasmonic acid accumulation in wounded *Arabidopsis*. *J Biol Chem* 284:34506–34513
- Goldman N, Yang Z (1994) A codon-based model of nucleotide substitution for protein-coding DNA sequences. *Mol Biol Evol* 11:725–736
- Harwood JL (2000) Eicosanoids and related compounds in plants and animals. In: AF Rowley, H. Kuhn and T. Schewe (Ed.). London, UK: Portland Press Ltd, *The New Phytologist*. 146:23–25
- Heitz T, Bergey DR, Ryan CA (1997) A gene encoding a chloroplast-targeted lipoxygenase in tomato leaves is transiently induced by wounding, systemin, and methyl jasmonate. *Plant Physiol* 114:1085–1093
- Hu B, Jin J, Guo A-Y, Zhang H, Luo J, Gao G (2015) GSDS 2.0: an upgraded gene feature visualization server. *Bioinformatics* 31:1296–1297
- Kumar S, Stecher G, Li M, Knyaz C, Tamura K (2018) MEGA X: molecular evolutionary genetics analysis across computing platforms. *Mol Biol Evol* 35:1547–1549
- Lescot M et al (2002) PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. *Nucleic Acids Res* 30:325–327
- Li H, Huang Y (2017) Expression of brown-midrib in a spontaneous sorghum mutant is linked to a 5'-UTR deletion in lignin biosynthesis gene *SbCAD2*. *Scientific Reports* 7:1–10
- Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the 2⁻ $\Delta\Delta$ CT method. *Methods* 25:402–408
- Ma F, Yang X, Shi Z, Miao X (2020) Novel crosstalk between ethylene- and jasmonic acid-pathway responses to a piercing-sucking insect in rice. *New Phytol* 225:474–487
- Melan MA, Dong X, Endara ME, Davis KR, Ausubel FM, Peterman TK (1993) An *Arabidopsis thaliana* lipoxygenase gene can be induced by pathogens, abscisic acid, and methyl jasmonate. *Plant Physiol* 101:441–450
- Mewis I, Tokuhisa JG, Schultz JC, Appel HM, Ulrichs C, Gershenzon J (2006) Gene expression and glucosinolate accumulation in *Arabidopsis thaliana* in response to generalist and specialist herbivores of different feeding guilds and the role of defense signaling pathways. *Phytochemistry* 67:2450–2462
- Moran PJ, Thompson GA (2001) Molecular responses to aphid feeding in *Arabidopsis* in relation to plant defense pathways. *Plant Physiol* 125:1074–1085
- Nalam VJ, Keeretaweep J, Sarowar S, Shah J (2012) Root-derived oxylipins promote green peach aphid performance on *Arabidopsis* foliage. *Plant Cell* 24:1643–1653
- Nalam VJ, Keeretaweep J, Shah J (2013) The green peach aphid, *Myzus persicae*, acquires a LIPOXYGENASE5-derived oxylipin from *Arabidopsis thaliana*, which promotes colonization of the host plant. *Plant Signaling Behavior* 8:e22735
- Ogunola OF et al (2017) Characterization of the maize lipoxygenase gene family in relation to aflatoxin accumulation resistance. *PLoS ONE* 12:e0181265
- Padilla MN, Hernández ML, Sanz C, Martínez-Rivas JM (2009) Functional characterization of two 13-lipoxygenase genes from olive fruit in relation to the biosynthesis of volatile compounds of virgin olive oil. *Journal of Agricultural and Food Chemistry* 57:9097–9107
- Padilla MN, Hernández ML, Sanz C, Martínez-Rivas JM (2012) Molecular cloning, functional characterization and transcriptional regulation of a 9-lipoxygenase gene from olive. *Phytochemistry* 74:58–68
- Quevillon E, Silventoinen V, Pillai S, Harte N, Mulder N, Apweiler R, Lopez R (2005) InterProScan: protein domains identifier. *Nucleic Acids Res* 33:W116–W120
- Reymond P, Weber H, Damond M, Farmer EE (2000) Differential gene expression in response to mechanical wounding and insect feeding in *Arabidopsis*. *Plant Cell* 12:707–719
- Royo J, Vancanneyt G, Pérez AG, Sanz C, Störmann K, Rosahl S, Sánchez-Serrano JJ (1996) Characterization of three potato lipoxygenases with distinct enzymatic activities and different organ-specific and wound-regulated expression patterns. *J Biol Chem* 271:21012–21019
- Sarde SJ, Kumar A, Remme RN, Dicke M (2018) Genome-wide identification, classification and expression of lipoxygenase gene family in pepper. *Plant Mol Biol* 98:375–387
- Schommer C et al (2008) Control of jasmonate biosynthesis and senescence by miR319 targets. *PLoS Biol* 6:e230
- Sharp AJ et al (2005) Segmental duplications and copy-number variation in the human genome. *The American Journal of Human Genetics* 77:78–88
- Shen J et al (2014) A 13-lipoxygenase, *TomloxC*, is essential for synthesis of C5 flavour volatiles in tomato. *J Exp Bot* 65:419–428
- Shivaji R et al (2010) Plants on constant alert: elevated levels of jasmonic acid and jasmonate-induced transcripts in caterpillar-resistant maize. *J Chem Ecol* 36:179–191
- Siedow JN (1991) Plant lipoxygenase: structure and function. *Annu Rev Plant Biol* 42:145–188
- Singh B, Padmaja P, Seetharama N (2004) Biology and management of the sugarcane aphid, *Melanaphis sacchari* (Zehntner) (Homoptera: Aphididae), in sorghum: a review. *Crop Protection* 23:739–755
- Song H, Wang P, Li C, Han S, Lopez-Baltazar J, Zhang X, Wang X (2016) Identification of lipoxygenase (LOX) genes from legumes and their responses in wild type and cultivated peanut upon *Aspergillus flavus* infection. *Scientific reports* 6:35245
- Steczek J, Donoho GP, Clemens JC, Dixon JE, Axelrod B (1992) Conserved histidine residues in soybean lipoxygenase: functional consequences of their replacement. *Biochemistry* 31:4053–4057
- Team RC (2013) R: A language and environment for statistical computing. Austria, Vienna
- Thaler JS, Humphrey PT, Whiteman NK (2012) Evolution of jasmonate and salicylate signal crosstalk. *Trends Plant Sci* 17:260–270
- Tzin V et al (2015) Dynamic maize responses to aphid feeding are revealed by a time series of transcriptomic and metabolomic assays. *Plant Physiol* 169:1727–1743
- ul Hassan MN, Zainal Z, Ismail I, (2015) Green leaf volatiles: biosynthesis, biological functions and their applications in biotechnology. *Plant Biotechnol J* 13:727–739

- Umate P (2011) Genome-wide analysis of lipoxygenase gene family in Arabidopsis and rice. *Plant Signaling Behavior* 6:335–338
- Upadhyay RK, Mattoo AK (2018) Genome-wide identification of tomato (*Solanum lycopersicum* L.) lipoxygenases coupled with expression profiles during plant development and in response to methyl-jasmonate and wounding. *J Plant Physiol* 231:318–328
- USDA N (2017) Quick stats USDA National Agricultural Statistics Service, Washington, DC <https://quickstats.nass.usda.gov/>. Accessed 3 May 2019
- Wang JW, Wu JY (2005) Nitric oxide is involved in methyl jasmonate-induced defense responses and secondary metabolism activities of *Taxus* cells. *Plant Cell Physiol* 46:923–930
- Wasternack C, Parthier B (1997) Jasmonate-signalled plant gene expression. *Trends Plant Sci* 2:302–307
- Woldemariam MG, Ahern K, Jander G, Tzin V (2018) A role for 9-lipoxygenases in maize defense against insect herbivory. *Plant Signaling Behavior* 13:4709–4723
- Yan L et al (2013) Role of tomato lipoxygenase D in wound-induced jasmonate biosynthesis and plant immunity to insect herbivores. *PLoS Genet* 9:e1003964
- Yang F et al (2015) Analysis of key genes of jasmonic acid mediated signal pathway for defense against insect damages by comparative transcriptome sequencing. *Scientific Reports* 5:16500
- Yu C-S, Cheng C-W, Su W-C, Chang K-C, Huang S-W, Hwang J-K, Lu C-H (2014) CELLO2GO: a web server for protein subCELLular LOcalization prediction with functional gene ontology annotation. *PLoS ONE* 9:e99368
- Zarate SI, Kempema LA, Walling LL (2007) Silverleaf whitefly induces salicylic acid defenses and suppresses effectual jasmonic acid defenses. *Plant Physiol* 143:866–875
- Zhang B, Chen K, Bowen J, Allan A, Espley R, Karunairetnam S, Ferguson I (2006) Differential expression within the LOX gene family in ripening kiwifruit. *J Exp Bot* 57:3825–3836
- Zhang P-J, He Y-C, Zhao C, Ye Z-H, Yu X-P (2018) Jasmonic acid-dependent defenses play a key role in defending tomato against *Bemisia tabaci* nymphs, but not adults. *Frontiers in plant science* 9:1065
- Zhou G, Qi J, Ren N, Cheng J, Erb M, Mao B, Lou Y (2009) Silencing OsHI-LOX makes rice more susceptible to chewing herbivores, but enhances resistance to a phloem feeder. *Plant J* 60:638–648
- Zhou G et al (2014) The 9-lipoxygenase Osr9-LOX1 interacts with the 13-lipoxygenase-mediated pathway to regulate resistance to chewing and piercing-sucking herbivores in rice. *Physiol Plant* 152:59–69
- Zhu-Salzman K, Salzman RA, Ahn J-E, Koiwa H (2004) Transcriptional regulation of sorghum defense determinants against a phloem-feeding aphid. *Plant Physiol* 134:420–431
- Zuckerkindl E, Pauling L (1965) Evolutionary divergence and convergence in proteins. In: Bryson V, Vogel HJ (eds) *Evolving genes and proteins*. Elsevier, Amsterdam, pp 97–166

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